

YOLO-Ball: Real-time tennis ball detection under occlusion and motion blur

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Abstract

This paper proposes YOLO-Ball, an improved real-time detection model addressing tennis ball detection challenges including small target size, motion blur, and player/racket occlusion. The main contributions include: (1) A multi-branch occlusion-aware attention mechanism for dynamic multi-scale feature fusion, (2) A dual-flow shallow fusion pyramid combining P2 features with bidirectional fusion to enhance small target and blur handling, and (3) A dynamic balance loss integrating Normalized Gaussian Wasserstein Distance (NWD) and Intersection over Union (IoU) with learnable alignment weights. Experiments on the constructed tennis dataset show 82.2% precision and a mean Average Precision at an Intersection over Union threshold of 0.5 (mAP@0.5) of 70.9%, outperforming YOLOv8/v10/v11 by up to 12.5%. YOLO-Ball also demonstrates generalization for volleyball and football detection, thereby providing a robust detection framework for scenarios involving high-speed, small targets.

Keywords

tennis ball detection, occlusion-aware attention, multi-scale feature fusion, real-time object detection, motion blur analysis, bidirectional feature pyramid, dynamic balance loss, normalized Gaussian Wasserstein distance, sports match analysis, small target detection

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Introduction

In the context of contemporary sporting events, artificial intelligence (AI) has become a prevalent technology in sports such as soccer, basketball, and rugby. For instance, Semi-Automated Offside Technology (SAOT) systems in soccer and optical tracking technologies in the National Basketball Association (NBA) utilize AI-driven computer vision to assist referees and provide real-time tactical data.¹

Despite these advancements, the reliable detection of tennis balls remains a persistent challenge due to the sport's unique dynamics. First, tennis balls occupy a negligible portion of the visual field and are easily obscured by background noise.² Secondly, ball velocities exceeding 200 km/h frequently induce motion blur and drag shadows, impeding feature extraction.³ Furthermore, frequent occlusions by players and rackets introduce complex interference patterns.⁴

To overcome these difficulties in professional tournaments, high-end hardware solutions have been widely adopted. The Hawk-Eye system, for example, utilizes multi-camera arrays to achieve millimeter-level

accuracy through 3D trajectory reconstruction.⁵ While effective in mitigating occlusion, its prohibitive cost and complex calibration requirements restrict its use to elite competitions.⁶ In contrast, this study targets low-cost, amateur, and youth training scenarios, aiming to develop a monocular vision method that balances high precision with real-time performance on standard hardware.

Although deep learning-based single-view methods offer deployability advantages, balancing accuracy and speed remains difficult. Traditional two-stage models like Faster R-CNN incur high computational latency (> 2.3 s),^{7,8} while transformers like RT-DETR^{9,10} often struggle to meet high frame-rate requirements. Standard YOLO variants (v8/v10/v11), though faster,

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Figure 1. Sample images exhibiting motion blur, occlusion, and complex scenarios.

often yield suboptimal precision for small, fast-moving targets.^{11,12}

Recent research has focused on optimizing detection heads and feature fusion architecture to address these limitations. Regarding feature fusion, researchers have sought to enhance cross-scale information flow. For example, RepGFPN¹³ and BiFPN¹⁴ reconstruct multi-scale interactions to improve layer fusion. Similarly, methods employing bidirectional connections¹⁵ and ternary feature encoders¹⁶ have been shown to better balance local details with global semantic information. Simultaneously, attention mechanisms and head optimization have been explored to handle scale variations and occlusion. Techniques such as the Scale-Aware Attention (SAA) module¹⁷ and Task-Aligned Dynamic Heads¹⁸ effectively suppress noise and emphasize small targets. Furthermore, incorporating 3D attention weights¹⁹ or coordinate attention²⁰ has proven useful in sharpening occluded features. However, purely enhancing features often neglects the alignment issues inherent in small-target loss functions. While recent works have proposed geometric optimizations like SIOU²¹ or MPD-IOU,²² dynamic balancing between feature alignment and positional regression in blurred scenarios remains underexplored.

To address the challenges of detecting tennis balls including occlusion, small target size, and motion blur, this paper proposes YOLO-Ball, a model with three integrated innovations. It employs an occlusion-aware attention module that dynamically fuses multi-scale features of obscured regions while suppressing background interference through adaptive weight adjustment. To better handle small-target detection and motion blur, we employ a bidirectional feature pyramid. This pyramid incorporates multi-branch auxiliary fusion to maintain representational flexibility without compromising inference speed. Finally, a dynamic balance loss combines IoU with Normalized Gaussian Wasserstein Distance, automatically optimizing position and feature alignment through learnable parameters to improve localization precision in blurred frames.

Materials and methods

Dataset

A total of 198 tennis match videos (720P, 16:9 aspect ratio) were collected from public tournament websites.

As this study utilized publicly available broadcast footage, specific ethical approval for human subjects was not required.

From these match videos, 4428 images containing tennis balls in flight were randomly sampled. Each image was proportionally scaled and padded to a resolution of 640×640 to form the experimental dataset. All tennis balls were manually annotated using LabelMe annotation tool (MIT Computer Science and Artificial Intelligence Laboratory, MA, USA) with rectangular bounding boxes. The dataset was partitioned into training, validation, and test sets in a ratio of 7:2:1.

Based on visual attributes, the images were categorized into four groups: 2412 (54.5%) images with fully visible tennis balls, 705 (15.9%) with motion blur, 562 (12.7%) with partial occlusion, and 749 (16.9%) with both blur and occlusion. As illustrated in Figure 1, accurate detection remained challenging due to the small pixel area of the balls (typically 0.02%–0.15% of the total image area), motion blur, and frequent partial occlusion by players or rackets.

Performance evaluation metrics

To evaluate the performance of YOLO-Ball in the task of tennis ball detection, this study employed precision, recall, and mean Average Precision (mAP) as the core evaluation metrics. Precision measured the accuracy of the model's predictions, representing the proportion of correctly detected tennis balls among all detected objects; a high precision indicated a lower rate of false positives. Recall assesses the model's ability to identify all actual tennis balls, with a high recall signifying fewer missed detections. The mean Average Precision (mAP), calculated as the area under the precision-recall curve, provides a comprehensive performance measure.

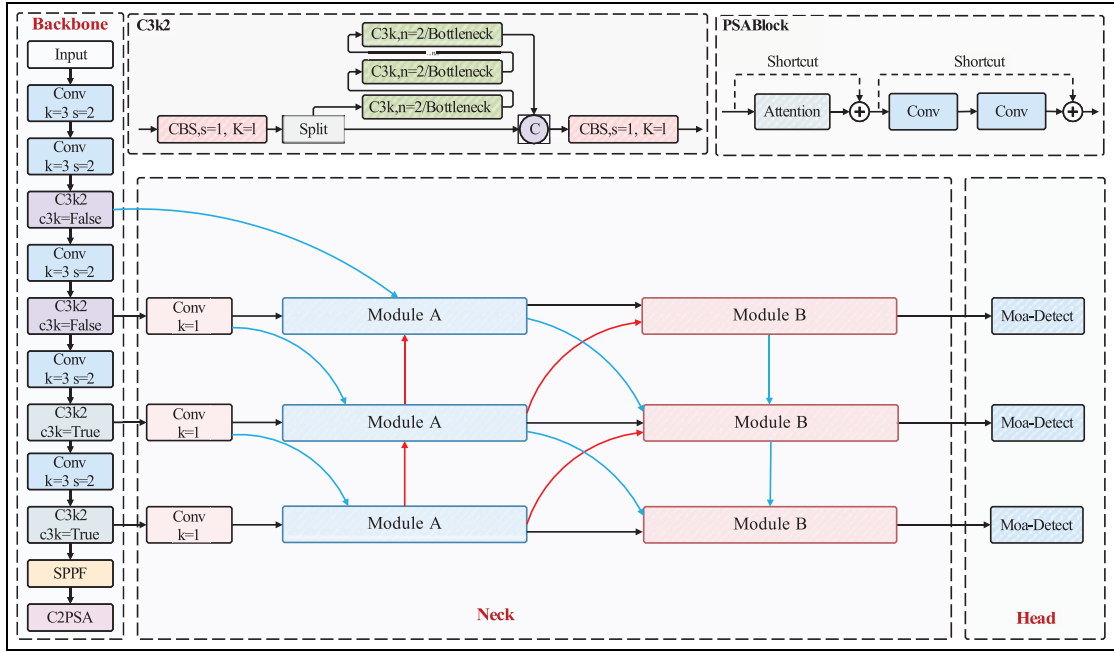


Figure 2. YOLO-Ball network architecture.

YOLO-Ball network architecture

YOLO-Ball is a real-time tennis ball detection model, the structure of which is shown in Figure 2. The model was divided into three parts: the backbone, the neck, and the head. These parts are responsible for feature extraction, feature fusion, and target localization, respectively.

The backbone consisted of a series of convolutional and C3k2 modules.²³ Convolution was primarily employed for feature extraction, whereas C3k2 carried out separation, convolution, and fusion operations. To enhance spatial awareness and global contextual modeling, the C2PSA module was introduced²³ at the end of the backbone. It integrated coordinate information into feature maps and employed point-line self-attention to capture long-range dependencies, enhancing the model's perception of target geometry and complex spatial relationships.

The neck of the proposed model consisted of Dual-Flow Shallow Fusion Pyramid (DSFP). The DSFP utilized the different scale features P2, P3, P4 and P5 as inputs, employing a bidirectional feature fusion from the top-down and from the bottom-up to enhance the model's capacity to detect small-scale and rapidly moving targets. In the initial bottom-up pathway, the Shallow-Layer Assisted Fusion module (SAF) is responsible for extracting multiscale features from the backbone and performing initial assisted fusion at the shallow network layers. Concurrently, the Deep-Layer Assisted Fusion module (DAF) gathers gradient information from each layer via denser connections in the second top-down pathway, thereby guiding the head to generate output features across three resolutions.

The proposed model addressed frequent occlusion from rackets and players by introducing the MOA-detect head, which employs a multi-branch attention mechanism to reweight feature maps and enhance the detection of occluded tennis balls.

Dual-flow shallow fusion pyramid

The low pixel proportion of tennis balls makes detection using only the high-level feature maps (P3–P5) from the backbone. To enhance the detection of targets exhibiting large frame-to-frame displacement, the high-resolution P2 layer was incorporated and construct a shallow-assisted bidirectional pyramid module termed DSFP. The DSFP module took multi-scale features (P2: $160 \times 160 \times C$, P3: $80 \times 80 \times C$, P4: $40 \times 40 \times C$, P5: $20 \times 20 \times C$) as input. Additionally, depthwise separable convolution is employed to minimize computational overhead (Figure 3).

The individual feature maps of the inputs are processed through convolutional layers and the C3k2 module for lateral feature enhancement. The features enhanced by the C3k2 module are subjected to multi-scale feature fusion in both top-down and bottom-up directions. In the bottom-up path: The model undergoes a process of downsampling and fusion from P2 to P5 from the bottom up. The C3k2 module downsamples the high-resolution features using stride-2 convolutions, and then fuses them with neighboring high-level semantics through concatenation followed by 1×1 convolution. Then in the top-down path the model is upsampled and fused from the P5 to the P2 level, from top to bottom. The C3k2 module enhances the high-

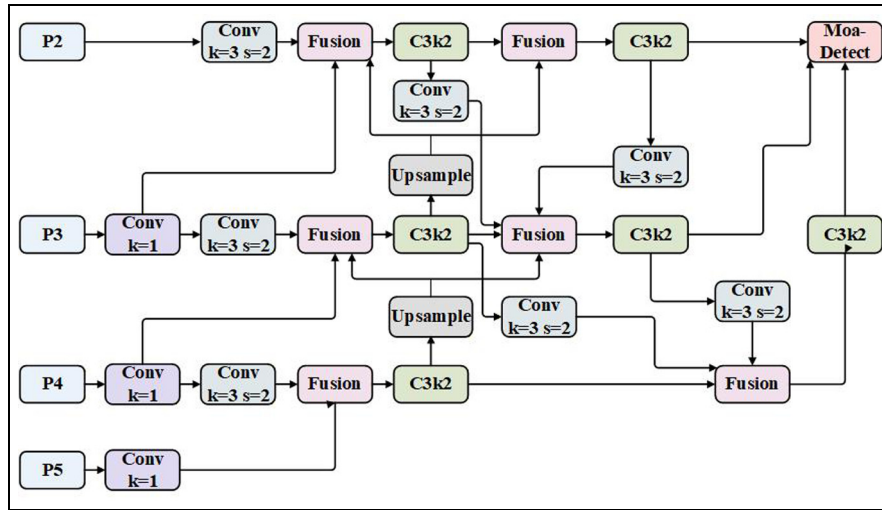


Figure 3. Dual-flow shallow fusion pyramid architecture.

level semantics features, which are then fused with the underlying information from the neighboring levels via bilinear upsampling and element-wise summation.

Multi-scale occlusion attention detect

To enhance the model's capacity to detect the occluded target and address the intricate scenario wherein the tennis balls are obscured by players,²⁴ rackets, and other objects, it was necessary to implement a series of modifications. Separated and Enhancement Attention was carried out to construct the Multi-Scale Occlusion Attention Module (MOA), as illustrated in Figure 4. This module was applied this module to the model's detection head.

The MOA module performed multi-scale feature extraction through three branches of a parallel DCovN structure, see dashed boxes in Figure 4. Each DCovN employs a set of convolutional kernels of varying sizes, resulting in a hierarchically distributed receptive field. The employment of small convolutional kernels is indicative of a focus on local details, while medium kernels are employed to capture partial occlusion relations. Conversely, large kernels are responsible for extracting global context, thereby facilitating the inference of features pertaining to the occluded region. The multi-branching structure of the model was designed to enhance its capacity to detect both minute and occluded targets.

The configuration of DCovN is illustrated in Figure 4(b), where the input feature map is initially divided into non-overlapping patches, that is, the feature map is initially subdivided into multiple patches of equivalent size and mapped to the high-dimensional feature space. The subsequent stacking of multiple depthwise convolution and pointwise convolution modules is employed for the purpose of feature extraction. In this process,

depthwise convolution operates on each channel information independently, while pointwise convolution facilitates cross-channel feature fusion. In the subsequent stage of the process, the global average pooling is performed on each DCovN output and the original feature maps. The results of this process are then fused and sent to the fully connected layer, with the aim of computing each channel's attention. The difference in the channel's attention weight is then amplified by means of an exponential operation, and the attention weight is finally applied to the feature map adjustment, to emphasize the features of occluded targets.

Dynamic balance loss

Tennis balls occupy few pixels in tennis match videos, and a small change in the position of the tennis ball detection box may lead to a huge change in IoU, making it easy for positive and negative samples to have similar IoU values and difficult for the network to converge.²⁵ Given the circular nature of the tennis ball, modeling the axis-aligned bounding box with a 2D Gaussian distribution to construct probability isocontour ellipses can more accurately represent the detection target. Therefore, the introduction of therefore propose introducing the Normalized Gaussian Wasserstein Distance (NGWD)²⁵ loss to optimize the model. The method combines Gaussian distribution models for predicted and ground-truth boxes, using Wasserstein distance to improve feature alignment.

Similarity measures based on Wasserstein distance. Let the mean and variance of the positions of the Predicted Box and Ground Truth Box be (μ_P, σ_P) and (μ_Q, σ_Q) , respectively, where μ denotes the center position of the box, and σ denotes the scale (width and height) of the box. The Wasserstein distance between the

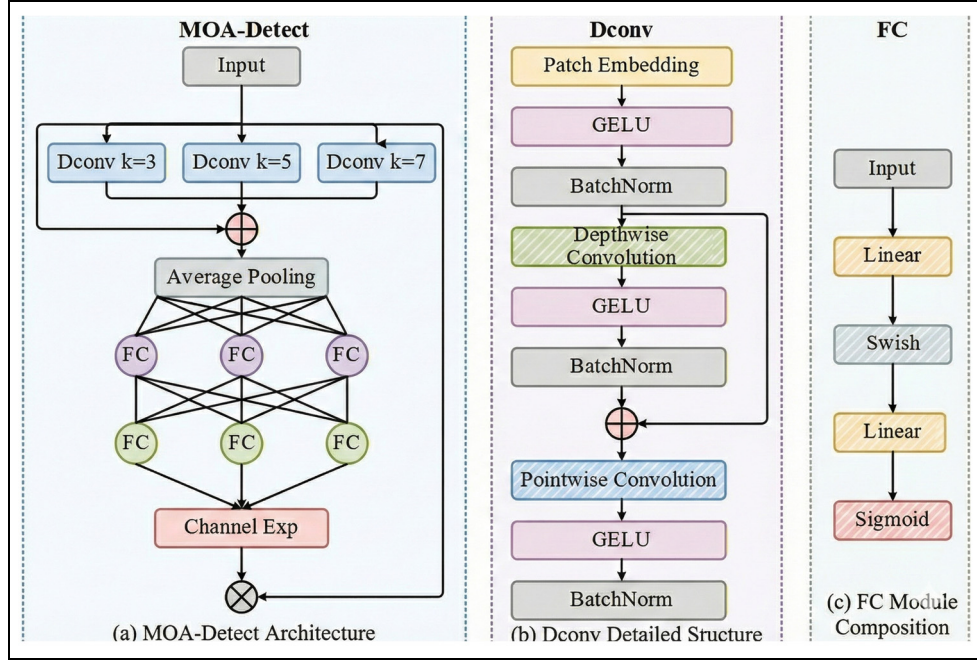


Figure 4 Multi-scale occlusion attention detect: (a) Overall architecture of MOA-Detect head; (b) Detailed structure of the Dconv module; and (c) Detailed structure of the FC module.

Predicted Box and Ground Truth Box under Gaussian distribution is:

$$W_2^2(P, Q) = \|\mu_P - \mu_Q\|_2^2 + \|\sigma_P - \sigma_Q\|_2^2 \quad (1)$$

$\|\mu_P - \mu_Q\|_2^2$ denotes the distance between the centroids of the two frames, and $\|\sigma_P - \sigma_Q\|_2^2$ denotes the scale difference of the frames. The Wasserstein distance not only quantifies the deviation of the position, but also focuses on the scale matching of the frames, and thus enables better alignment of small targets and subtle features. Further to accommodate target detection at different scales, the Wasserstein distance is normalized:

$$NGWD(P, Q) = \frac{W_2^2(P, Q)}{\sigma_P + \sigma_Q} \quad (2)$$

where $\sigma_P + \sigma_Q$ are normalization factors that balance the differences between targets at different scales.

Combined loss function of NGWD and IoU. Combining NGWD with IoU, the weights of both are dynamically adjusted by learnable parameters to achieve a balance between feature alignment and positional alignment.

$$LOSS = \alpha \cdot IoU + 1 - \alpha \cdot NGWD \quad (3)$$

Where α is a learnable parameter that automatically balances the influence of IoU and NGWD. In scenarios with cluttered backgrounds or severe occlusion, the model reduces IoU's weight and increases NGWD's contribution, leveraging NGWD's superior sensitivity

to feature-space alignment. Conversely, in simple scenes, IoU retains dominance for efficient positional calibration.

Compared to the conventional IoU loss, the proposed method, which combines NGWD and IoU, addresses several challenges in small target detection. These include spatial feature alignment, multi-scale target adaptation, and dynamic adjustment in complex scenes. NGWD employs Gaussian distribution modeling, enabling predicted box and ground truth box to be aligned more accurately in the feature space, a process that is effective in capturing fine features in small target detection. Furthermore, the incorporation of a normalization factor enables the NGWD to adaptively adjust the influence of targets at varying scales, thereby enhancing the model's performance in scenarios involving high-speed motion and complex occlusion. The combination of IoU and NGWD facilitates dynamic adjustment of the weighting of the two by a learnable weighting parameter. This dynamic adjustment enables the model to adapt to the detection needs of different scenarios and improves robustness in complex background and occlusion situations.

Results and analysis

Improved feature fusion pyramid

Table 1 presents the comparative performance of the proposed Dual-Flow Shallow Fusion Pyramid (DSFP) against state-of-the-art feature fusion modules, including GFPN,¹⁵ GOLD-YOLO,²⁶ HS-FPN,²⁷ SDI,²⁸ and BiFPN.²⁹ The evaluation focused on parameter

Table 1. Comparative analysis of feature fusion methods.

Model	Params (M)	FLOPs (G)	Precision (%)	mAP@0.5
GFPN	3.66	8.2	78.3	65.1
GOLD-YOLO	5.90	6.2	72.1	65.4
HS-FPN	1.82	5.6	77.5	66.2
SDI	2.63	6.7	76.3	67.3
BiFPN	1.92	6.3	78.0	66.2
DSFP (Ours)	1.83	5.8	78.8	68.2

Bold values indicate the best performance for each metric.

efficiency (Params), computational cost (FLOPs), and detection accuracy (Precision and mAP@0.5).

The DSFP module was efficient, requiring only 1.83M parameters and 5.8G FLOPs (Table 1), a configuration that surpassed the parameters and FLOPs requirements of other modules while maintaining low computational overhead. The DSFP module achieves a detection precision of 78.8%, and a mAP@0.5 of 68.2%. In contrast, GFPN can also achieve 78.3% detection precision, but the number of parameters and computation are 100% and 41% higher than DSFP, respectively. GOLD-YOLO exhibits the lowest performance among the compared models. BiFPN demonstrates a high degree of similarity to DSFP with regard to the number of parameters and computations. Nevertheless, its detection precision remains somewhat inferior to that of DSFP, as evidenced by mAP@0.5. Furthermore, HS-FPN and SDI demonstrate commendable performance metrics in terms of minimal parameter requirements and mAP@0.5. And DSFP achieves a superior balance between model complexity and detection capability. Compared to HS-FPN which has a similar number of parameters, DSFP enhances precision and mAP@0.5 by 1.3% and 2.0%, respectively. This finding suggests that DSFP can more effectively balance image detail features and deep semantic features during the bidirectional flow of features. This balanced feature representation results in an improvement in the perception of small-sized and fast-moving targets.

Incorporate Attention Mechanism

To enhance occlusion handling, we introduce the Multi-Scale Occlusion Attention (MOA) mechanism into the detection head. MOA is compared compare MOA with various attention modules, including BAMBlock, CoordAtt, LSKA, LSKBlock, MLCA, MPCA, SEAttention, SimAM, and so on, through a series of comparative experiments. The results of these experiments are presented in Table 2. MOA achieved 79.5% precision, while mAP@0.5 reaches 69.7%, both of which are higher than those of other attention modules.

To further analyze the role of attention mechanisms in tennis ball detection, the heat map visualization is used to explore the complex scenes such as small

Table 2. Comparative analysis of attention mechanisms.

Attention module	Precision (%)	Recall (%)	mAP@0.5
BAMBlock	78.4	68.1	67.0
CoordAtt	78.3	69.2	68.9
LSKA	79.1	67.0	68.2
LSKBlock	79.3	69.9	66.8
MLCA	78.3	72.7	70.7
MPCA	78.3	70.5	67.4
TripletAttention	73.5	70.0	66.7
SEAttention	78.2	69.7	69.0
SimAM	78.6	67.3	68.2
MOA (Ours)	79.5	69.9	69.7

Bold values indicate the best performance for each metric.

targets, drag shadows and occlusion, and the results are shown in Figure 5.

In the region affected by occlusion and drag shadows, MOA can focus on the key objects with the help of multi-scale feature fusion and dynamic weight adjustment. The high activation regions such as red and yellow are distributed in the heatmap, which can capture the subject features more accurately while suppressing the background interference. In contrast, other attention mechanisms have a certain degree of activation of tennis targets, but they often suffer from diffuse attention and an inability to focus on key target features. The color distribution of attention mechanisms such as BAMBlock, CoordAttention, SEAttention, etc. is more dispersed, and the red high-activation areas are less concentrated, making it difficult to accurately focus on the core features of the tennis ball. SimAM, SEAttention, TripletAttention, LSKBlock, MLCA, etc. Some of the attention mechanisms have more than one high-hotness region, which leads to the dispersed focus of the model and fails to highlight the main body of the tennis ball effectively. MPCA, MLCA, LSKA, etc. include the background region in the scope of attention, and the generated high-hotness region is too large, which interferes with the accurate extraction of the target features of the tennis ball.

Loss Function Comparison Experiment

To verify the effectiveness of NGWD's loss function with dynamic IoU adjustment, it was compared with

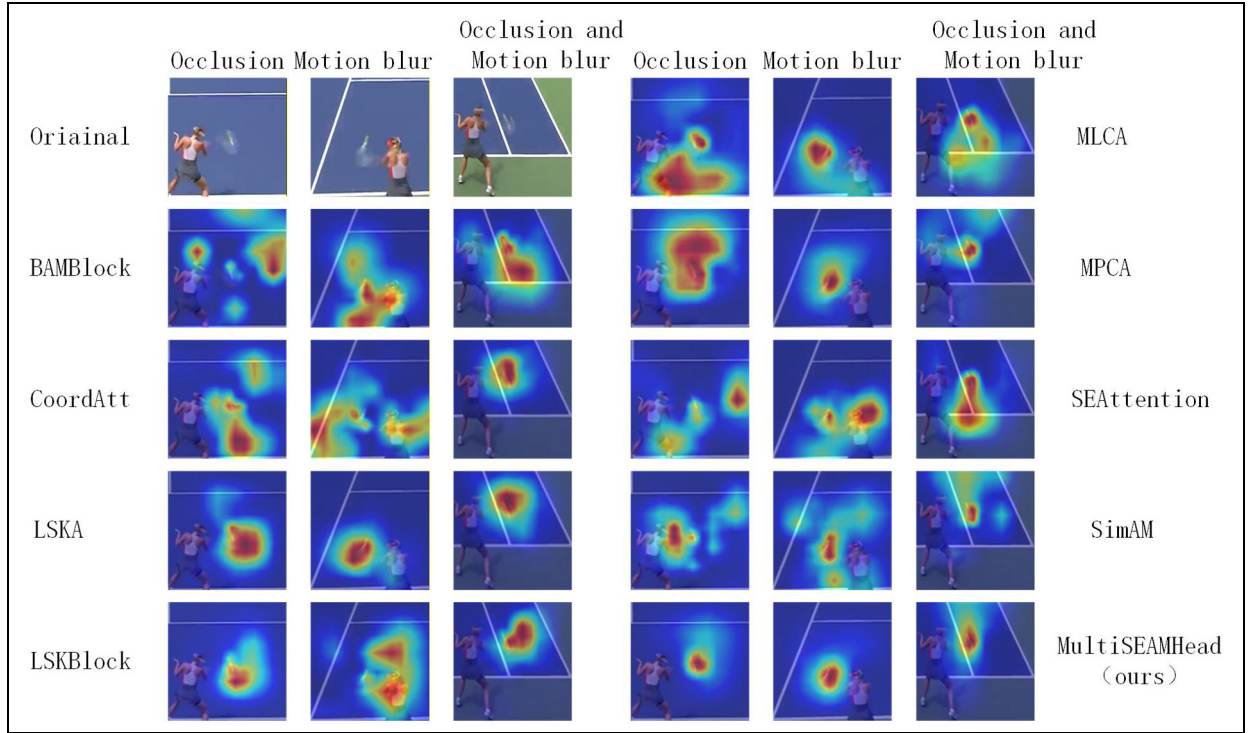


Figure 5. Comparison of heat maps corresponding to different attention mechanisms.

Table 3. Comparative study of loss functions.

Loss function	Precision (%)	Recall (%)	mAP@0.5
WIoU	74.1	63.6	65.9
NWD	77.4	64.1	65.2
WIoU (0.5) + NWD (0.5)	76.0	69.9	66.8
WIoU (0.2) + NWD (0.8)	76.3	70.7	67.4
DBL (Ours)	79.5	68.5	68.4

Bold values indicate the best performance for each metric.

various loss functions, and the results are shown in Table 3.

The tennis ball detection precision is found to be 77.4% when using NWD alone, but the mAP@0.5 is only 65.2%. When NWD and WIoU are combined with fixed weights ($\alpha = 0.5$), the mAP@0.5 improves,

but the model's detection precision decreases, indicating that static weight allocation struggles to balance feature and positional alignment. By contrast, the proposed dynamic balanced loss function adapts the weight ratio of NWD and WIoU using learnable parameters. This adaptive strategy retains NWD's ability to model the feature distribution of small targets while inheriting WIoU's precise bounding box regression constraints. The model precision reaches 79.5%, with a mAP@0.5 of 68.4%.

Ablation study

Through ablation experiments, the contributions of the dual-flow shallow fusion pyramid, multi-scale occlusion attention detect, and the loss function of the dynamic balance loss to the YOLO-Ball detection performance

Table 4. Ablation study of model components.

Components			FLOPs (G)	Params (M)	Precision (%)	Recall (%)	mAP @0.5
MOA	DSFP	DBL					
×	×	×	6.4	2.47	74.1	63.6	65.9
✓			6.1	3.07	79.5	69.9	69.7
	✓		6.3	1.83	78.8	68.1	68.2
		✓	6.3	2.46	76.3	69.0	66.8
✓	✓		6.0	2.46	81.2	70.9	70.4
✓		✓	6.1	3.07	79.7	68.4	68.2
✓	✓	✓	6.0	2.46	82.2	71.6	70.9

Table 5. Performance comparison under different conditions.

Condition	Precision (%)	Recall (%)	mAP@0.5
Fully visible	87.6	75.4	76.4
Motion blur	78.9	67.2	72.1
Occlusion	76.7	66.4	68.4
Motion blur and occlusion	72.2	62.9	63.0

are verified, respectively, and the results are shown in Table 4.

As shown in Table 4, the baseline model achieves 74.1% precision, 63.6% recall, and 65.9% mAP@0.5 with 6.4G FLOPs and 2.47M parameters. Adding only the MOA-Head reduces FLOPs to 6.1G but increases parameters to 3.07M, while improving precision and mAP@0.5 to 79.5% and 69.7%, respectively. Incorporating the DSFP module further optimizes the model to 6G FLOPs and 2.46M parameters, boosting precision to 81.2% and mAP@0.5 to 70.4%. Finally, when MOA-Head, DSFP, and DBL are integrated, the model reaches its optimal performance of 82.2% precision, 71.6% recall, and 70.9% mAP@0.5, demonstrating a clear synergistic effect that enhances capability in handling small and occluded targets.

Evaluation of detection accuracy under different interference factors

Supplementary experiments were conducted supplementary experiments to quantify the impact of key challenges by partitioning the test set (1443 images) into subsets based on occlusion (575 images), motion blur (603 images), and clarity (265 images).

As shown in Table 5, the model achieves its best performance (87.6% precision) under the “Fully Visible” condition. However, a performance drop was observed when motion blur or occlusion is introduced, with the most substantial degradation occurring when both challenges are present (72.2% precision). These results confirm that small target size, occlusion, and motion blur are all critical factors leading to performance degradation. While the proposed method exhibits robustness in mitigating occlusion effects, the precision results are not driven by occlusion alone, as motion blur presents a comparable, if not greater, challenge.

Comparison with existing relevant studies

Table 6 compares the performance of YOLO-Ball with other mainstream target detection models on the tennis dataset. The YOLO-Ball model achieved a precision of 82.2% and mAP@0.5 of 70.9%, surpassing the performance of general-purpose detectors such as YOLOv8, YOLOv10, and RT-DETR. Specifically, compared to YOLOv11, the proposed model improved precision by 8.1% and mAP@0.5 by 5.0%, demonstrating superior

Table 6. Performance comparison of tennis dataset.

Model	Precision (%)	Recall (%)	mAP@0.5
Faster R-CNN	21.2	22.3	8.1
RT-DETR	71.5	68.0	60.7
YOLOv8	72.4	66.4	61.3
YOLOv10	69.7	61.5	58.8
YOLOv11	74.1	63.6	65.9
YOLO-Ball (Ours)	82.2	71.6	70.9

Bold values indicate the best performance for each metric.

capability in handling the specific challenges of tennis match scenarios.

Comparison of different ball datasets

To evaluate the generalization capability of YOLO-Ball, it was tested it along with Faster R-CNN, RT-DETR, YOLOv8, YOLOv10, and YOLOv11 on the YOLOv5 ball detection dataset, which includes volleyball, soccer, and baseball videos. The dataset composition varies across ball types: the volleyball subset contains 33% blur and 17% occlusion; the soccer subset has 24% blur and 33% occlusion; and the baseball subset exhibits 41% blur and 7% occlusion. This diversity in testing conditions provides a robust basis for evaluating model performance across different scenarios. The comparative results are summarized in Table 7.

With regard to the task of detecting volleyballs, YOLO-Ball achieves optimal performance, exhibiting a precision of 94.9%, a recall of 87%, an mAP@0.5 of 92.8%, and YOLOv10 achieves 92.4% mAP@0.5. The YOLOv8 model, on the other hand, exhibits an mAP@0.5 of 91.4%, suggesting comparable efficacy in volleyball detection.

For soccer ball detection, YOLO-Ball achieved the highest mAP@0.5, with precision and recall reaching 57.4% and 45.3% respectively, outperforming all other models. The mAP@0.5 for YOLOv10 and YOLOv11 is 33.2% and 39.9%, respectively, while Faster R-CNN exhibits a lower performance, achieving only 0.29% mAP@0.5.

For baseball detection, YOLOv10 achieved the highest mAP@0.5, while YOLO-Ball attained a lower mAP@0.5 of 74.2% but demonstrated superior precision of 79.4%.

The experimental results show that for different target categories, there are differences in the detection ability of each model, but YOLO-Ball shows comprehensive performance with high generalization in all items.

Figure 6 illustrates the detection effects of various models under varying conditions. The higher performance in volleyball detection can be attributed to its larger relative size and lower occlusion frequency in the dataset, as analyzed in Section 3.6.

Even so, volleyballs still occupied a limited pixel area (typically less than 0.15% of the frame) and are subject

Table 7. Performance comparison of different models on sports ball detection(%).

Models	Volleyball			Football			Baseball		
	Precision	Recall	mAP@0.5	Precision	Recall	mAP@0.5	Precision	Recall	mAP@0.5
Faster R-CNN	23.2	16.4	8.4	5.6	1.2	0.3	15.6	6.5	2.1
RT-DETR	74.2	62.5	37.2	45.6	38.7	34.8	73.0	70.2	72.1
YOLOv8	94.0	84.0	91.4	43.4	39.4	38.6	72.8	73.7	76.3
YOLOv10	93.9	85.2	92.4	51.1	29.1	33.2	75.3	71.0	77.9
YOLOv11	92.5	84.0	90.0	46.2	21.4	39.9	74.5	70.5	71.1
YOLO-Ball (Ours)	94.9	87.0	92.8	57.4	45.3	43.9	79.4	69.6	74.2

Bold values indicate the best performance for each metric.

**Figure 6.** Comparison with mainstream target detection models on different ball datasets.

to partial occlusions, indicating the model's ability to generalize across different object properties.

In soccer ball detection, for instance, partial occlusion caused by the league emblem led to missed detections by RT-DETR, YOLOv10, and YOLOv11, while YOLOv8 achieved detection but with low confidence (0.27). In contrast, YOLO-Ball identified the target with a higher confidence score of 0.45, demonstrating robustness in complex scenarios involving partial occlusion, motion blur, and scale variation.

Discussion

Feasibility for real-world tennis analysis

The experimental results indicated that YOLO-Ball achieved 82.2% Precision and 70.9% mAP@0.5. These metrics demonstrate the model's feasibility for practical use in tennis tracking-driven quantitative analysis, particularly in low-cost application scenarios such as training for amateurs and youth players. It is important to note that there is no universally defined

accuracy threshold for confidently applying tracking technology in practice; the required level of precision is dependent on the specific application context. For instance, the current performance level is sufficient to provide valuable reference for general trajectory analysis and statistical purposes during training sessions. However, for strict line-judging assistance or fine-grained match analysis, higher accuracy may be necessary.

Robustness under partial occlusion

A key challenge identified in this study was detection under occlusion. As observed in the soccer ball experiments (Figure 6), our model successfully maintained tracking even when the target was partially obscured by overlay logos. It is crucial to clarify that our evaluation focused on *partial occlusion*. For frames where the ball was partially hidden by overlays or players, the ground truth was annotated based on the visible portions of the object. The successful detection in these scenarios confirms that the proposed occlusion-aware attention mechanism (MOA) effectively extracts features from the remaining visible regions, rather than relying on hallucinating completely invisible targets.

Trade-off between 3D estimation and real-time performance

While our experimental results demonstrate that YOLO-Ball achieves high precision in 2D detection, we acknowledge that full 3D trajectory reconstruction remains a desirable goal for professional analysis. In the early stages of this research, we initially explored monocular 3D coordinate estimation (e.g. using improved VGG-based approaches). However, we observed that the substantial computational overhead of such methods conflicted with our primary goal of real-time, lightweight deployment. Consequently, we prioritized optimizing the model for robust 2D tracking to ensure accessibility for low-cost hardware setups.

Practical applications of 2D tracking

Although lacking depth of information, the robust 2D tracking achieved by YOLO-Ball offers practical value. This study focuses on 2D tracking because the output data can be directly utilized for essential match analytics. Specifically, the generated 2D trajectory data effectively supports a variety of practical applications, including shot placement distribution analysis, approximate ball speed estimation, and rally statistics. This practical utility makes the proposed solution highly suitable for amateur enthusiasts and youth training scenarios where expensive 3D systems are impractical.

Future directions

While the proposed method demonstrates promising results, there remain avenues for further improvement. Future work will focus on optimizing inference speed for mobile platforms to facilitate edge deployment. Additionally, plans are in place to extend the model's application to other ball sports, such as table tennis or badminton, to validate its generalization capabilities in diverse athletic scenarios.

Conclusion

Detecting tennis balls in match footage presents persistent challenges, primarily due to their small size, frequent occlusion, and motion blur. YOLO-Ball was proposed these issues, we proposed YOLO-Ball, a lightweight algorithm designed to enhance detection performance in such complex scenarios. The contributions are summarized as follows: (1) A multi-branch occlusion-aware attention mechanism was introduced to dynamically fuse features and suppress background interference. (2) A shallow-assisted bidirectional feature pyramid was designed to incorporate high-resolution P2 features, improving small-target detection. (3) A dynamic balance loss function was proposed to integrate NGWD and IoU with learnable weights, optimizing alignment for blurred targets. In summary, YOLO-Ball provides a robust framework for real-time sports analysis, demonstrating significant potential for assisting amateur and youth training.

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Appendix

List of notation

C	Number of channels
k	Kernel size
IoU	Intersection over Union
$NGWD$	Normalized Gaussian Wasserstein Distance
P	Predicted bounding box
Q	Ground truth bounding box
s	Stride
W_2	Wasserstein distance
α	Weighting parameter for the combined loss function
μ	Center position of the bounding box
σ	Scale (width and height) of the bounding box